

PERFORMANCE ANALYSIS OF INGAAS-BASED DOUBLE GATE MOSFETS: IMPACT OF GATE DIELECTRIC MATERIAL AND THICKNESS

Technical Project Document & Abstract

Architecture:	Symmetric Double-Gate (DG) MOSFET
Simulation Platform:	Synopsys Sentaurus TCAD
Channel Materials:	Silicon (Si) vs. Indium Gallium Arsenide (InGaAs)

1. The Engineering Problem

As conventional silicon-based MOSFETs scale down into the deep nanometer regime, they face severe physical limitations such as aggressive short-channel effects (SCEs), high gate leakage current, and a substantial degradation of electrostatic control over the channel region. These structural and electrostatic bottlenecks limit further improvements in integrated circuit operational processing speed and power efficiency, driving the vital engineering need to explore alternative high-mobility compound materials and multi-gate device architectures.

2. Proposed Solution & Methodology

This project investigates a combined structural and material engineering solution by performing a rigorous performance evaluation of a symmetric Double Gate (DG) MOSFET configuration. Using physics-based Synopsys Sentaurus TCAD simulations, a comprehensive comparative analysis is carried out between two distinct channel material setups: traditional Silicon (Si) and Indium Gallium Arsenide (**InGaAs**), an advanced high-mobility III-V compound semiconductor. The presence of dual gates on both sides of a ultra-thin channel significantly reduces short-channel variations by improving gate-to-channel capacitive coupling.

3. Parametric Investigations

To identify optimized low-power configurations, systematic parameter variation sweeps were conducted on the device gate stacks within the Technology Computer-Aided Design (TCAD) environment:

- **Oxide Scaling:** Systematically varying and analyzing the direct impact of physical gate oxide thickness (T_{ox}) variations ranging from 1 nm to 4 nm.
- **Dielectric Engineering:** Evaluating critical electrostatic coupling enhancements when substituting conventional Silicon Dioxide (SiO_2) with alternative high-k dielectric materials, including Silicon Nitride (Si_3N_4), Aluminum Oxide (Al_2O_3), and Hafnium Dioxide (HfO_2).

4. Key Performance Metrics Evaluated

The electrical performance and operational logic parameters of the structures were rigorously extracted from simulated DC transfer characteristics (I_{DS} - V_{GS}) to evaluate:

- Threshold Voltage (V_T) roll-off and inversion launch requirements.
- Subthreshold Swing (SS) performance in millivolts per decade (mV/dec) to track subthreshold power efficiency.
- ON-state drive current (I_{ON}) capacity versus OFF-state leakage current (I_{OFF}) dynamics.

5. Technical Findings & Conclusions

The simulation data clearly demonstrate that the *InGaAs*-based DG MOSFET delivers a significantly higher ON-state drive current (reaching order magnitudes of 10^{-4} A) and an exceptionally sharp, efficient switching transition compared to the silicon baseline configuration due to its superior electron mobility and low effective mass. Furthermore, minimizing the physical oxide thickness (T_{ox}) enhances gate control, leading to a diminished subthreshold swing, while the implementation of alternative high-k dielectrics like *HfO₂* maximizes gate capacitance without violating structural physical limit thresholds. This combined material and geometric optimization enables high-performance, ultra-scalable, and low-power architectures perfectly suited for next-generation nanoscale electronics.